

Predictability-Aware Soft Set Hopping Against Reactive Jamming

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Abstract—Hard recent-set exclusion in sparse multicarrier frequency hopping creates a predictable boundary spike: $C_{L_h+1} = S/(N - L_h S)$ grows with memory depth, and probability-aware exposure equals this height. We propose predictability-aware soft hopping with capped inclusion probabilities (PASH-cap), using soft penalties and a probability cap that bounds worst-case exposure. On a coded OFDM grid, PASH-cap reduces BLER from 0.138 (MCSH-L2) to 0.020 at delay three; against a causal attacker with 12 candidate lags, BLER is 0.045 versus 0.126 for MCSH-L2, with BLER 0.039–0.047 under degraded feedback. Five online-learning variants all sustain BLER 0.454–1.000.

Index Terms—Anti-jamming communications, frequency hopping, OFDM, reactive jamming, predictability, boundary spike, probability cap.

I. INTRODUCTION

FREQUENCY-HOPPING (FH) links evade reactive jammers by varying their occupied frequencies, but a history-aware jammer can exploit structure in the hopping rule itself. This problem is practically relevant: in tactical and vehicular networks, a sparse multicarrier waveform must maintain connectivity under a follower jammer that tracks spectral occupancy, and set-valued hopping is a natural fit for these bandwidth-efficient links. Recent defenses encompass index-modulated FH [1], data-driven and reinforcement learning (RL) hopping [2]–[6], anti-jamming games [7], and jamming-modulation schemes [8]. Detection-focused work addresses reactive jammer identification in OFDM [9], [10] and co-operative sensing [11]. Sequence constructions provide wide-gap or no-hit-zone properties [12], [13], while coded OFDM studies emphasize receiver-side mitigation [14]–[18]. These approaches broadly share a design philosophy of *avoiding* recent frequency usage, whether through combinatorial sequence constraints or learned policies.

This letter examines a different question: when recent frequency sets are excluded too aggressively, what structural consequence does that exclusion itself create? Most existing work assumes that stronger avoidance of recent usage monotonically improves anti-jamming performance. We show that this assumption fails in a precise and quantifiable way. Hard exclusion guarantees zero collision with a follower whose delay falls within the protected memory, but it also shrinks the conditional choice space, making frequencies just beyond that memory disproportionately likely to reappear. We prove that the collision probability at the first unprotected lag is $C_{L_h+1} = S/(N - L_h S)$, which grows toward unity as the protected memory deepens. This is not an engineering tradeoff that could be eliminated by a better hard-exclusion design; it is a structural consequence of the zero-collision guarantee itself (we refer to this hard memory-constrained set hopping scheme as MCSH throughout). Moreover, the same concentration makes the conditional inclusion probabilities maximally

informative to a probability-aware attacker: the candidate-aware exposure equals the boundary-spike height, so deeper memory simultaneously raises the worst-case predictability cost.

This diagnostic result distinguishes our work from the existing literature. Wide-gap and no-hit-zone constructions provide deterministic pairwise collision-avoidance at specified delays [12], [13], but they target FH patterns with one active subcarrier per symbol ($S = 1$) and do not address the conditional-inclusion concentration that arises under set-valued hopping with $S > 1$ active subcarriers. RL-based policies can learn to avoid recent frequencies through online collision feedback [2], [3], but they offer no interpretable bound on the exposure they create, and their closed-loop nature introduces a vulnerability to feedback manipulation that we show empirically. Rather than optimizing a hopping policy, we diagnose a structural consequence of memory and provide a tunable open-loop alternative that avoids the convergence pitfalls of online learning.

Building on this diagnosis, we propose predictability-aware soft hopping with capped inclusion probabilities (PASH-cap). It replaces hard exclusion with exponential penalties on recent subcarrier use and enforces a per-subcarrier probability cap $p_{\max}$ through systematic sampling. The cap directly bounds the worst-case conditional exposure: $R_{\text{cand}} \leq p_{\max}$, in contrast to MCSH where $R_{\text{cand}} = S/(N - L_h S)$ is unbounded. PASH-cap is open-loop (requires no jammer feedback), incurs no learning latency, and cannot be manipulated through corrupted or falsified collision feedback.

This letter makes three contributions. First, we derive the collision at the first unprotected lag of MCSH and prove that the boundary-spike magnitude $S/(N - L_h S)$ follows unavoidably from the zero-collision guarantee; deeper memory amplifies rather than removes the exposure. Second, we propose PASH-cap with an explicit inclusion-probability cap, implemented by a fixed-size sampler that preserves target marginals exactly. Third, we evaluate five methods spanning the key design dimensions of open-loop set-valued FH (Random, BCD, LFSR, MCSH, PASH-cap) under fixed-delay, probability-aware, and feedback-aided causal attacks, plus a 5-variant $\times$ 4-threat online-learning matrix confirming that closed-loop strategies without anti-predictability control fail systematically.

II. SYSTEM AND THREAT MODEL

In symbol m , the transmitter activates $\mathcal{A}_m \subseteq \{1, \dots, N\}$, where $|\mathcal{A}_m| = S$. A jammer occupies an equal-size set $\mathcal{J}_m$, and the normalized collision is

$$c_m = \frac{|\mathcal{A}_m \cap \mathcal{J}_m|}{S}. \quad (1)$$

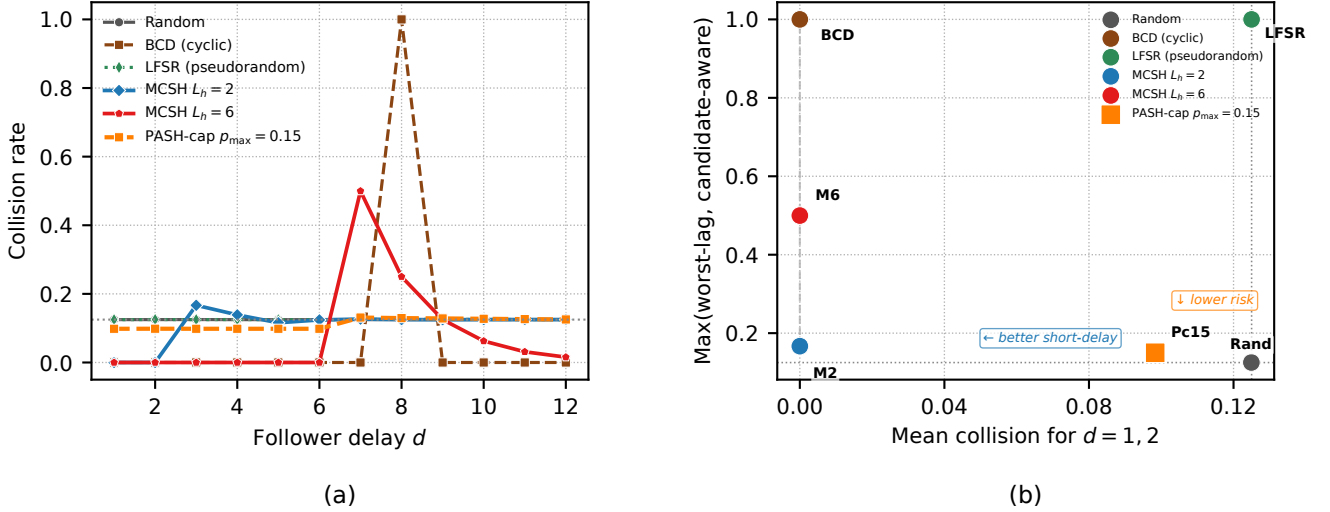


Fig. 1. Structural exposure. (a) MCSH suppresses protected lags but produces the boundary spike in (3); PASH-cap limits concentration. (b) Short-delay protection and worst conditional exposure form a tradeoff. BCD and LFSR are deterministic stress cases ($R_{\text{cand}}=1.0$).

Define the lag- d self-collision $C_d = \mathbb{E}[|\mathcal{A}_m \cap \mathcal{A}_{m-d}|/S]$. A fixed-delay follower uses $\mathcal{J}_m = \mathcal{A}_{m-d}$, so its expected collision equals C_d . We test $d = 1$ and $d = 3$. Two deterministic constructions are included as structural stress cases that span the design space: a block-cyclic deterministic construction (BCD) that cycles through N/S orthogonal sets ($C_1=\dots=C_7=0$, $C_8=1.0$), and a linear-feedback-shift-register (LFSR) based pseudorandom construction representing the classical FH strategy of tactical systems ($C_d \approx S/N$, but $R_{\text{cand}}=1.0$ because the conditional marginals remain deterministic, $p_{k,m} \in \{0, 1\}$).

The probability-aware attacker knows the conditional marginals $p_{k,m} = \Pr(k \in \mathcal{A}_m | \mathcal{A}_{<m})$ and jams the S bins with the largest $p_{k,m}$.

The feedback-aided causal attacker maintains exponentially weighted moving average (EMA) collision scores for $d \in \{1, \dots, 12\}$, selects a delay via ϵ -greedy ($\epsilon=0.10$), and jams the corresponding historical set. It observes the normalized collision only after acting, and never accesses $\mathcal{A}_m$ when choosing an action.

III. BOUNDARY EXPOSURE AND PASH-CAP

MCSH with hard memory L_h selects S bins uniformly outside the union of the L_h most recent active sets whenever feasible. It guarantees

$$\mathcal{A}_m \cap \mathcal{A}_{m-d} = \emptyset, \quad 1 \leq d \leq L_h. \quad (2)$$

Proposition 1 (Boundary spike): If $(L_h + 1)S \leq N$, the collision at the first unprotected lag is

$$C_{L_h+1} = \frac{S}{N - L_h S}. \quad (3)$$

Proof: By definition, $\mathcal{A}_m$ is disjoint from each of the L_h most recent active sets. Under the MCSH rule, each of these L_h sets avoids all predecessors within its own L_h -step window, so the L_h sets that $\mathcal{A}_m$ excludes are themselves pairwise disjoint. Together they occupy exactly $L_h S$ distinct bins, leaving $N - L_h S$ unprotected bins from which $\mathcal{A}_m$ draws

uniformly. The historical set $\mathcal{A}_{m-L_h-1}$ is at lag L_h+1 from $\mathcal{A}_m$ and is therefore outside $\mathcal{A}_m$'s exclusion window. Moreover, $\mathcal{A}_{m-L_h-1}$ is disjoint from all of $\mathcal{A}_{m-L_h}, \dots, \mathcal{A}_{m-1}$ (each of these symbols excludes it through their own L_h -step memory). Hence all S of its elements lie in the unprotected complement. Each such element has probability $S/(N - L_h S)$ of appearing in the independently drawn $\mathcal{A}_m$, giving

$$C_{L_h+1} = \frac{1}{S} \cdot S \cdot \frac{S}{N - L_h S} = \frac{S}{N - L_h S},$$

establishing (3). The expression is strictly increasing in L_h : deeper memory amplifies the very exposure it defers. $\square$

Proposition 2 (Unavoidable predictability cost): Under MCSH, the probability-aware exposure equals the boundary-spike height,

$$R_{\text{cand}} = C_{L_h+1} = \frac{S}{N - L_h S}. \quad (4)$$

Proof: Hard exclusion makes all bins in the unprotected complement equally likely to be selected, so the conditional inclusion probabilities $p_{k,m}$ are uniform over that complement (zero for excluded bins, uniform otherwise). The attacker jams the S bins with largest $p_{k,m}$, all of which lie in the unprotected complement. The expected fraction of active bins captured is therefore $S/(N - L_h S)$, matching (3). $\square$

Proposition 2 identifies a structural lower bound for hard-exclusion strategies: any rule that guarantees $C_d = 0$ for $d \leq L_h$ necessarily concentrates its conditional probability mass on at most $N - L_h S$ unprotected bins. By the pigeonhole principle, the top- S mass among these bins cannot have expected value below $S/(N - L_h S)$, and MCSH attains this lower bound by using the uniform distribution over the unprotected complement. Any other hard-exclusion rule that deviates from uniformity would suffer even higher R_{cand} . The designer thus faces an inescapable tradeoff: zero short-lag collision forces a lower bound on predictability, and this bound rises sharply as L_h increases, from 0.143 at $L_h=1$ to 0.500 at $L_h=6$, reaching 1.0 at $L_h=7$ (when $L_h S$ approaches

N). BCD, a block-cyclic deterministic construction, illustrates the extreme: perfect short-lag protection ($C_1 = \dots = C_7 = 0$) but $R_{\text{cand}} = 1.0$ because of its periodic structure (usage coefficient of variation (CV) of 0, max inclusion probability of 1.0). In contrast, PASH-cap's p_{max} bound is constant; it does not grow with H .

PASH-cap replaces hard exclusion with an exponential penalty on recent-use count. The exponential form keeps the weights strictly positive (avoiding hard zeros), uses a single parameter α , and makes contributions beyond H steps negligible without explicit truncation. With memory H , define

$$w_{k,m} = p_{\text{floor}} + \exp \left[-\alpha \sum_{\ell=1}^H \mathbf{1}\{k \in \mathcal{A}_{m-\ell}\} \right]. \quad (5)$$

The target marginal probabilities are obtained by capped normalization,

$$p_{k,m} = \min\{p_{\text{max}}, \eta_m w_{k,m}\}, \quad \sum_{k=1}^N p_{k,m} = S, \quad (6)$$

where η_m is found by bisection; a solution always exists because $\sum_k \min(p_{\text{max}}, \eta w_{k,m})$ is continuous in η with range $[0, Np_{\text{max}}] \supset [0, S]$. To realize these marginals exactly, PASH-cap uses fixed-size systematic sampling: bins are randomly permuted, a uniform offset is drawn, and bins whose cumulative probability intervals contain any of S equispaced thresholds are selected. The cap $p_{\text{max}} \leq 1$ ensures distinct bins, and exchangeability under the random permutation guarantees that bin k appears with probability $p_{k,m}$.

For any valid marginals, the probability-aware exposure

$$R_{\text{cand}} = \frac{1}{S} \sum_{k \in \text{TopS}(\mathbf{p}_m)} p_{k,m} \quad (7)$$

satisfies $S/N \leq R_{\text{cand}} \leq p_{\text{max}}$; Random attains the lower bound because all $p_{k,m} = S/N$.

Proposition 3 (PASH-cap guaranteed bounds): (i) *Candidate-aware exposure:* $R_{\text{cand}} \leq p_{\text{max}}$ for all m and all history realizations, independent of H and α . (ii) *Self-limiting feedback:* For any bin k with at least one appearance in the H -step window, its raw weight satisfies $w_{k,m} \leq p_{\text{floor}} + e^{-\alpha}$. With $\alpha = 1.4$, $e^{-\alpha} \approx 0.247$, so two or more appearances reduce the exponential contribution below 0.061. (iii) *Self-collision bound:* For any lag $d \geq 1$, the self-collision satisfies $C_d \leq p_{\text{max}}$.

Proof: (i) Each $p_{k,m} \leq p_{\text{max}}$ by (6), so the top- S sum is at most $S \cdot p_{\text{max}}$, yielding the bound. (ii) When $\sum_{\ell} \mathbf{1}\{k \in \mathcal{A}_{m-\ell}\} \geq 1$, the exponential term is at most $e^{-\alpha}$, so $w_{k,m} \leq p_{\text{floor}} + e^{-\alpha}$. With $\alpha = 1.4$, this is 0.447; two or more appearances drop it below 0.261. (iii) Let $\mathcal{F}_m$ denote the history $\mathcal{A}_{<m}$. The systematic sampling procedure draws a random permutation independent of $\mathcal{F}_m$ and selects bins whose cumulative probability intervals contain equispaced thresholds. Under the random permutation, each bin k is equally likely to occupy any position in the permuted order, and the threshold spacing guarantees $\Pr(k \in \mathcal{A}_m | \mathcal{F}_m) = p_{k,m}$ (a standard property of randomized systematic sampling; the

cap discussion following (6) provides the construction). For any lag $d \geq 1$,

$$\begin{aligned} \Pr(k \in \mathcal{A}_m \cap \mathcal{A}_{m-d}) &= \mathbb{E}[\mathbf{1}\{k \in \mathcal{A}_{m-d}\} \mathbb{E}[\mathbf{1}\{k \in \mathcal{A}_m\} | \mathcal{F}_m]] \\ &= \mathbb{E}[\mathbf{1}\{k \in \mathcal{A}_{m-d}\} p_{k,m}] \\ &\leq p_{\text{max}} \mathbb{E}[\mathbf{1}\{k \in \mathcal{A}_{m-d}\}]. \end{aligned} \quad (8)$$

By symmetry across bins, each bin has unconditional inclusion probability $\Pr(k \in \mathcal{A}_{m-d}) = S/N$, so $\mathbb{E}[\mathbf{1}\{k \in \mathcal{A}_{m-d}\}] = S/N$. Summing over all N bins and normalizing by S ,

$$C_d = \frac{1}{S} \sum_{k=1}^N \Pr(k \in \mathcal{A}_m \cap \mathcal{A}_{m-d}) \leq \frac{1}{S} \cdot N \cdot p_{\text{max}} \cdot \frac{S}{N} = p_{\text{max}}. \quad (9)$$

The bound requires no approximation and holds for every $d \geq 1$ under the PASH-cap sampling procedure. $\square$

Proposition 3 provides three guarantees that MCSH structurally cannot offer. Part (i) bounds the attacker's information advantage against the strongest probability-aware threat ($R_{\text{cand}} \leq p_{\text{max}}$); part (ii) explains the mechanism: a bin appearing even once in the recent window has its weight bounded by $p_{\text{floor}} + e^{-\alpha}$, preventing the probability concentration that creates MCSH's boundary spike; part (iii) bounds self-collision uniformly ($C_d \leq p_{\text{max}}$), which governs performance against fixed-delay followers ($d=1, 3$ in Table I). In contrast, MCSH's $C_{L_h+1} = S/(N - L_h S)$ grows toward unity as L_h increases. The bound in part (iii) is independent of d , H , and α : it follows solely from the exchangeability of the random permutation and the per-bin probability cap. Numerical validation confirms that PASH-cap's worst-case empirical C_d is 0.131 (bound 0.15); the bound is not tight but is uniform over all lags and independent of H and α . The per-bin usage CV is 0.031, consistent with long-run uniformity (versus 0.038 for Random on the same grid).

Remark (Steady-state behavior): The PASH-cap weight-and-sample procedure defines a Markov chain on the H -step history. Because the exponential-weight function and the random permutation in systematic sampling are both label-symmetric, and $p_{\text{floor}} > 0$ ensures that every active-set configuration has positive probability, the process tends toward an exchangeable steady state in which $\mathbb{E}[p_{k,m}] \approx S/N$ for every k . With the pointwise bound $p_{k,m} \leq p_{\text{max}}$ from (6), PASH-cap's per-bin probabilities are therefore concentrated in the narrow interval $[S/N, p_{\text{max}}] = [0.125, 0.15]$, independent of H . MCSH, by contrast, assigns probability 0 to $L_h S$ excluded bins and spreads the remaining mass over the unprotected complement, with its per-bin maximum $S/(N - L_h S)$ diverging toward 1.0 as $L_h \rightarrow N/S$. The mixing time is not characterized; the empirical usage CV of 0.031 (close to Random's 0.038, see below) indicates near-uniformity within the simulated horizon. This qualitative contrast explains PASH-cap's robustness against adaptive and probability-aware attackers.

We use $H = 6$, $\alpha = 1.4$, $p_{\text{floor}} = 0.20$, and $p_{\text{max}} = 0.15$. $H=6$ matches MCSH's $L_h=6$ for fair comparison; the exponential decay makes contributions beyond ~ 5 steps negligible ($e^{-7.0} \approx 9 \times 10^{-4}$). With $H=0$, weights are uniform and the cap is inactive ($S/N=0.125 < p_{\text{max}}$); the method reduces to Random. The per-symbol computation requires $O(NH)$ for weight accumulation and $O(N + S)$ for sampling. For the

tested configuration ($N=512$, $S=64$, $H=6$), approximately 3000 arithmetic operations per OFDM symbol are needed, which is negligible relative to typical cyclic-prefix durations.

A structural sweep over $\alpha \in [0.4, 2.0]$, $p_{\text{floor}} \in [0.10, 0.35]$, $p_{\text{max}} \in [0.12, 0.20]$ (180 grid points) selected ($\alpha=1.4$, $p_{\text{floor}}=0.20$, $p_{\text{max}}=0.14$); coded trials show BLER differs by < 0.005 across all threats from $p_{\text{max}}=0.15$, which we report for stronger short-delay protection. In the sweep, p_{max} was dominant (> 0.1 BLER shift); α and p_{floor} had smaller effects (< 0.02 each). At 9 dB SNR the ranking is preserved: PASH-cap vs. MCSH-L2 is 0.05 vs. 0.317 ($d=3$) and 0.057 vs. 0.293 (causal). A systematic comparison of five online-learning variants (ϵ -greedy at $\epsilon=0.10$ and 0.50, fast adaptation, entropy-regularized, UCB) across four threats confirms this risk: BLER ranges from 0.454 to 1.000, 8.6–23.7 $\times$ worse than Random. The candidate-aware attacker achieves BLER 1.000 against every variant because the learner’s objective (to minimize collision) drives it toward the same low-collision subcarrier set on every symbol. Once converged, the per-bin usage distribution becomes extremely concentrated (usage CV $\rightarrow 0$, analogous to BCD’s 0.000), making the conditional marginals deterministic from the attacker’s perspective. This is the structural opposite of PASH-cap, whose cap-enforced usage CV of 0.031 remains close to Random’s 0.038. Online learning without explicit anti-predictability control can be *worse* than no learning at all, regardless of exploration strategy.

IV. RESULTS

A. Protocol

All methods use $N = 512$, $S = 64$, QPSK, rate-1/2 K7 convolutional coding, mixture-LLR Viterbi receiver, AWGN at 12 dB SNR, 5 dB JSR, and 32-symbol frames. Coded trials run for at least 2000 frames, stopping after 100 frame errors or 5000 frames. Bootstrap intervals are descriptive.

B. Structural Mechanism

Fig. 1(a) confirms (3). MCSH with $L_h=2$ has $C_1=C_2=0$, $C_3 \approx 0.167$; $L_h=6$ gives $C_7=0.500$, matching Proposition 1. PASH-cap accepts $C_1 \approx 0.098$ but keeps its worst historical collision (0.131) and R_{cand} (0.150) near Random (0.125). Fig. 1(b) shows MCSH variants on a diagonal where lower short-delay protection forces higher worst-case risk; PASH-cap occupies a region inaccessible to hard exclusion.

C. Coded Reliability

Fig. 2(a) and Table I show that no rule dominates every threat. MCSH-L2 is strongest against $d=1$ (BLER 0.0010, hard exclusion eliminates collision) but its own BLER degrades $\sim 138\times$ from $d=1$ to $d=3$ (BLER 0.1375) due to the boundary spike. PASH-cap avoids this concentration, remaining at 0.0200. Against the causal attacker with 12 candidate lags, MCSH-L2 sees $> 80\%$ of actions concentrated on $d=3$ (BLER 0.1255), while PASH-cap’s lag distribution stays diffuse and its BLER of 0.0450 is near Random (0.0530).

The BLER gap between MCSH-L2 and PASH-cap at $d = 3$ (0.1375 vs. 0.0200, 6.9 $\times$) far exceeds their collision-rate

TABLE I
BLER COMPARISON ACROSS METHODS AND THREATS

Threat	Random	BCD	LFSR	MCSH-L2	PASH-cap
$d=1$	0.0425	0.0006	0.0690	0.0010	0.0172
$d=3$	0.0535	0.0060	0.0274	0.1375	0.0200
Prob.-aware	0.0422	1.000	1.000	0.1435	0.0870
Causal ($d \in \{1, \dots, 12\}$)	0.0530	1.000	0.0845	0.1255	0.0450

Bold = best per row. BCD and LFSR are structural stress cases ($R_{\text{cand}}=1.0$): any deterministic hopping rule is fully exploitable. PASH-cap meets the 100-error target in all rows. Full bootstrap 95% confidence intervals and supplementary results are available from the authors.

gap (0.167 vs. 0.098, 1.70 $\times$). This nonlinearity reflects the boundary spike creating collision bursts at lag L_h+1 within each frame; the rate-1/2 convolutional code ($K=7$) has finite memory, so a burst exceeding its correction capability causes frame loss even at moderate average rates. PASH-cap’s soft control distributes collisions uniformly. The collision-to-BLER mapping is governed by collision *concentration*, not average rate.

Under full probability awareness, Random achieves the theoretical lower bound (BLER 0.0422) because uniform $p_{k,m}$ offers no concentration to exploit. PASH-cap reaches 0.0870, substantially better than MCSH-L2 (0.1435).

BCD and an LFSR-based pseudorandom construction further confirm the phenomenon is not specific to MCSH: BCD has $C_1=\dots=C_7=0$, $C_8=1.0$, and BLER 1.000 against both adaptive and probability-aware attackers. LFSR, the classical FH strategy of tactical systems, has near-Random short-lag BLER (0.0690 at $d=1$, 0.0274 at $d=3$) and moderate causal-attacker BLER (0.0845), yet collapses to 1.000 against the probability-aware attacker because $R_{\text{cand}}=1.0$. Any deterministic mapping yields $p_{k,m} \in \{0, 1\}$ regardless of implementation, and is fully exploitable. Two alternative weighting schemes confirm the exponential form and cap are individually necessary: PASH-linear (linear decay) improves candidate-aware BLER to 0.066 but degrades $d=1$; PASH-nocap ($p_{\text{max}}=1$) improves $d=1$ to 0.006 but suffers candidate-aware BLER 0.354.

D. Feedback Robustness

Fig. 2(b) shows feedback degradation: MCSH-L2 assigns 80.0%–83.6% of actions to $d = 3$ under all conditions; the exposure is structural. PASH-cap keeps lag choices distributed (BLER 0.039–0.047).

E. Scope and Limitations

Table I spans the key design dimensions of open-loop set-valued FH: memoryless stochastic (Random), memoryless deterministic (BCD, LFSR), memory-based hard exclusion (MCSH), and memory-based soft exclusion (PASH-cap). BCD and LFSR represent deterministic FH constructions (whose behavior subsumes that of NHZ sequences naively adapted to $S>1$); the RL matrix represents closed-loop strategies. External baselines operate under different system models (wide-gap/NHZ: $S=1$; deep-RL: closed-loop state-action spaces) that preclude faithful reproduction. The comparison isolates

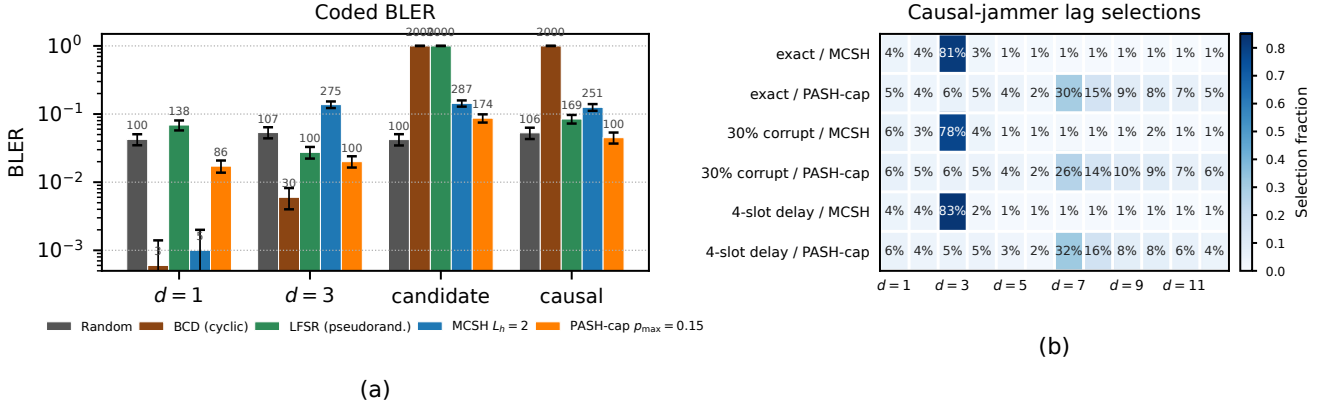


Fig. 2. Compact coded and feedback results. (a) BLER with descriptive bootstrap 95% intervals; numbers above bars are observed frame errors. (b) Causal-jammer lag-selection fractions under exact, corrupted, and delayed feedback; each row is normalized independently.

hopping-pattern exposure; it does not claim superiority over published methods. Attackers have strong information models; bootstrap intervals are descriptive. AWGN is assumed; fading remains future work.

A simple decision framework emerges: MCSH when the delay is short and known; PASH-cap when the delay is uncertain or the attacker adapts; Random when the attacker has full distributional knowledge. PASH-cap fills the region between these extremes.

V. CONCLUSION

Avoiding recently used frequencies in anti-jamming FH has an unintended structural consequence: it concentrates collision risk at the first unprotected lag. We proved the boundary spike $C_{L_h+1} = S/(N-L_hS)$ follows from the zero-collision guarantee (Proposition 2); deeper memory amplifies predictability risk. PASH-cap replaces hard exclusion with soft penalties and a per-subcarrier cap, providing $R_{\text{cand}} \leq p_{\max}$ (Proposition 3). The cap is essential: removing it raises candidate-aware BLER 4.1 $\times$. BCD and LFSR show the phenomenon is universal: any deterministic rule has $R_{\text{cand}} = 1.0$. Five online-learning variants all fail (BLER 0.454–1.000). Future work includes fading channels and hardware validation.

DATA AVAILABILITY

Code and data available from the authors.

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